

Coding & Solution

Date	26 June 2026
Team ID	SWIID-2026-1243
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Coding & Solution

The **EduGenie – Google Gemini Powered Learning Assistant** has been developed using a modular and scalable architecture. The backend is implemented with **FastAPI**, while the frontend is built using **HTML5, CSS3, and JavaScript**. Google **Gemini 1.5 Pro** is integrated to provide intelligent educational assistance, including question answering, concept explanations, quiz generation, text summarization, and personalized learning recommendations. The code follows clean coding practices, modular design, reusable components, proper error handling, and version control, ensuring maintainability and future scalability.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/nitishsoni07-ui/EduGenie-Google-Gemini-Powered-Learning-Assistant
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	FastAPI, Google Gemini 1.5 Pro API, Transformers, Jinja2
Key Features Implemented	

	Intelligent Question Answering, Concept Explanation, AI Quiz Generation, Educational Text Summarization, Personalized Learning Recommendations, Responsive User Interface, FastAPI REST APIs
Pending / Incomplete Features	User Authentication, Learning Progress Dashboard, User Profile Management, Database Integration (Future Enhancement)
Setup / Run Instructions	Clone the repository, install dependencies using pip install -r requirements.txt, configure the Gemini API key in the .env file, and start the application using uvicorn main:app --reload.



Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows a clean and modular architecture by separating the frontend, backend, AI services, and utility modules. FastAPI manages REST APIs, while Google Gemini 1.5 Pro powers the AI features. The repository is maintained using GitHub, allowing efficient collaboration, version control, and future feature enhancements. The current implementation satisfies the core functional requirements of the EduGenie learning assistant and is designed to be scalable for future improvements.